Which Learner Should We Use to Estimate Heterogeneous Treatment Effects?

IS 507 Midterm Presentation

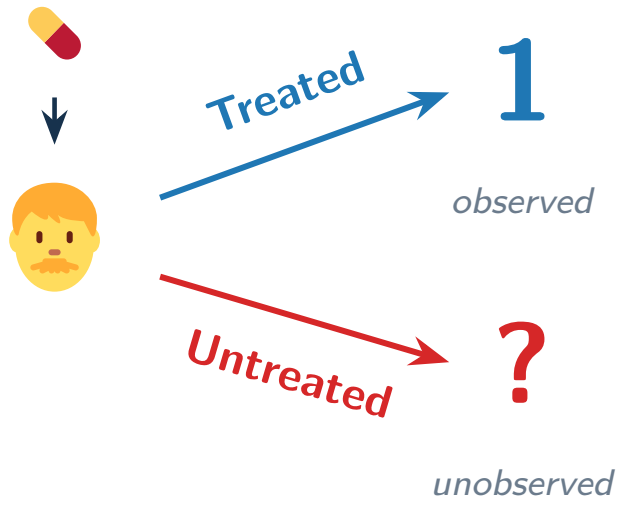
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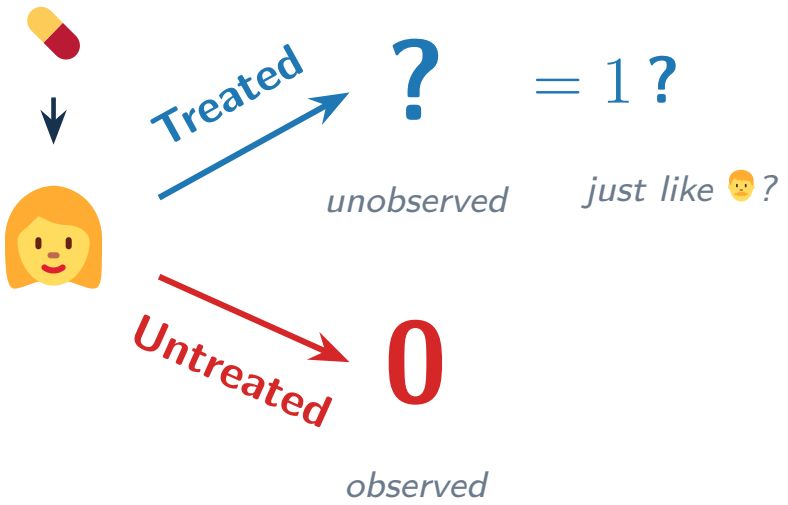
Based on Curth and van der Schaar, *Nonparametric Estimation of Heterogeneous Treatment Effects: From Theory to Learning Algorithms.* AISTATS 2021.

Motivation

Problem 1. We only see one outcome



Problem 2. Same effect for everyone?



What are we interested in?

$$\textbf{Causal effect} = \mathbb{E}\Big[\textbf{Treated outcome} - \textbf{Untreated outcome} \,\Big|\, \underbrace{\text{👤}}_{\textit{gender, age, ...}}\Big]$$

Motivation

1. Several **learners** for estimating causal effects

   ex) S-learner, T-learner, RA-learner, PW-learner, DR-learner, ...

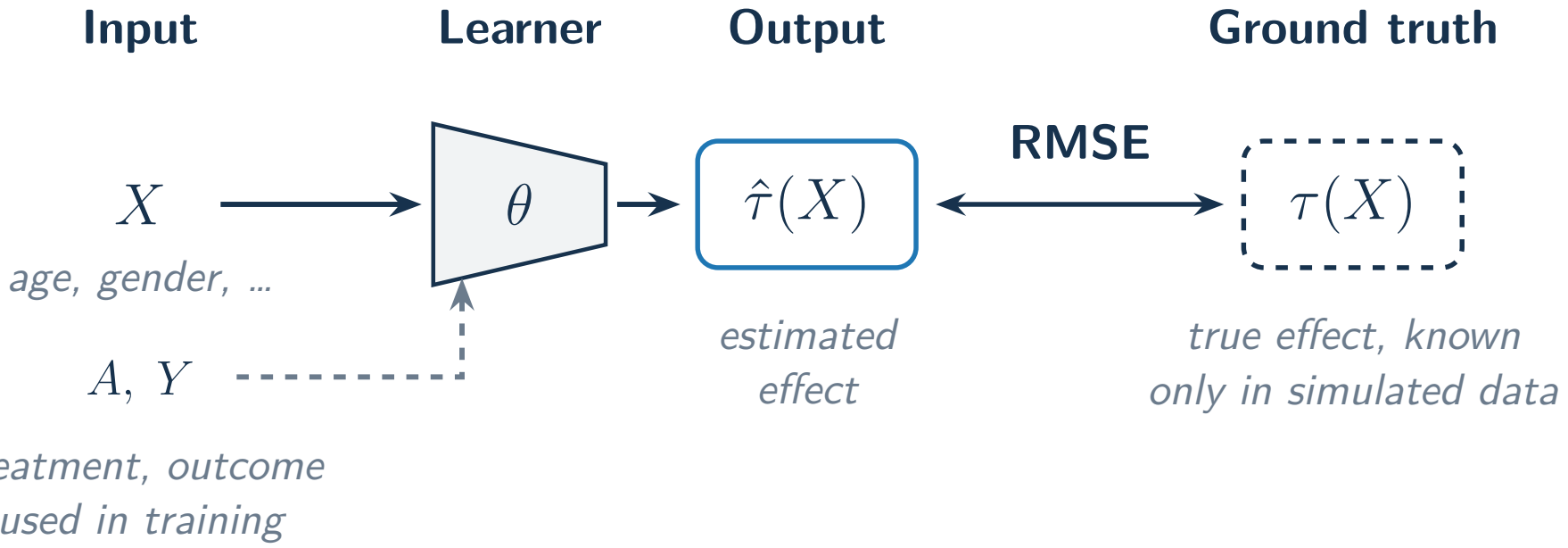
2. Q. In **which setting**, **which learner** is appropriate?

   ex) sample size, number of features, confounding, ...

Goal

Estimate causal effect under various settings

$$\tau(X) = \mathbb{E}\Big[\text{Treated outcome} - \text{Untreated outcome} \,\Big|\, X\Big] \qquad \text{👤} = X$$



Baseline Simulation Setup

What did the original paper investigate?

Why simulation? The true $\tau(X)$ is known $\Rightarrow$ the **Error** can be measured.

Property	**Specification**
Data	25 Gaussian covariates X, treatment $A \in \{0, 1\}$, outcome Y
Outcome	$\mu_0(X) = \mathbb{E}[Y \mid X, A = 0]$, $\mu_1(X) = \mathbb{E}[Y \mid X, A = 1]$ $\Rightarrow \ \tau(X) = \mu_1(X) - \mu_0(X)$
Sample size	Train: 1k / 2k / 5k / 10k Test: 500
Settings	(i) No effect: $\mu_0 = \mu_1$ (ii) Effect: $\mu_1 = \mu_0 + \tau$ (iii) μ_0 and μ_1 depend on different covariates

Limitations and Our Extensions

What remains insufficiently understood, and what do we change?

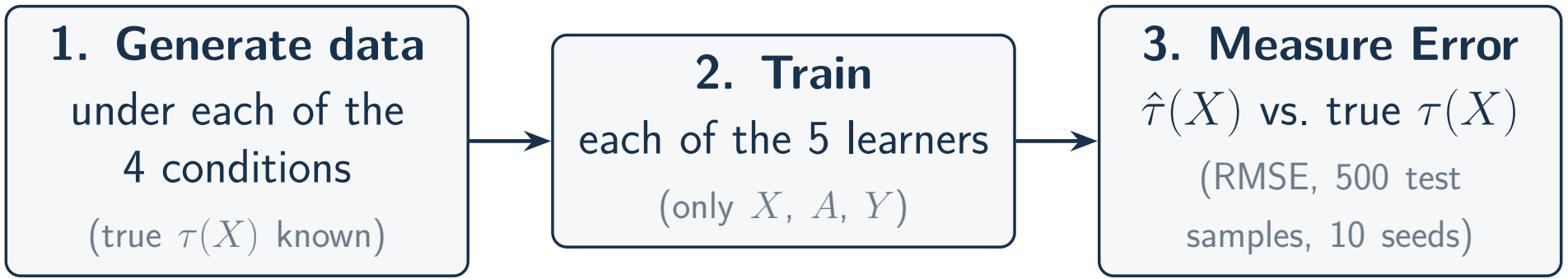
	Original paper	**Our extension**
Sample size	1k or more	Small: under 1k
Covariates	X: fixed at 25	X: more than 25
Overlap	Not varied	Strong vs. weak
Outcome shape	μ_0, μ_1: all quadratic	Linear / quadratic / nonlinear, set separately for μ_0 and μ_1

Overlap: every kind of X appears in both groups ($A = 1$ and $A = 0$).

$\Rightarrow$ Which learner stays **reliable** as conditions get harder?

Models and Training Plan

Learners Plug-in: S, T Two-step: RA, PW, DR



Objective How much does each learner's Error grow under the 4 conditions?

Conclusion

- The best learner for estimating $\tau(X)$ **depends on the setting**.
- The original paper compared learners in **simple and ideal** simulated settings.
- We test them under **more challenging** conditions: small samples, more covariates, weak overlap, complex μ_0, μ_1.
- **Next:** run the experiments and report the Error of each learner under each condition.

References

1. Curth, A., and van der Schaar, M. (2021). *Nonparametric Estimation of Heterogeneous Treatment Effects: From Theory to Learning Algorithms.* AISTATS, PMLR 130, 1810–1818.
2. Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). *Metalearners for Estimating Heterogeneous Treatment Effects Using Machine Learning.* Proceedings of the National Academy of Sciences, 116(10), 4156–4165.
3. Kennedy, E. H. (2023). *Towards Optimal Doubly Robust Estimation of Heterogeneous Causal Effects.* Electronic Journal of Statistics, 17(2), 3008–3049.
4. Jung, Y. (2025). *A Short Note on Empirical Excess Risk for Orthogonal Losses.* Technical Note. `https://yonghanjung.me/blog/2025/orthogonal-statistical-learning/`.